

# Demo: Loading Macros via include-in-header

This page demonstrates the `include-in-header` strategy for loading macros. The YAML front matter of this file includes:

```
format:
  html:
    include-in-header:
      - macros-header.html    # MathJax macro config script
  pdf:
    include-in-header:
      - macros.qmd            # raw LaTeX goes directly into preamble
  revealjs:
    output-file: demo-include-in-header-slides.html
    scrollable: true
    center: false
    include-in-header:
      - macros-header.html    # loader that reads macros.qmd
  docx: default
```

**Note (HTML/RevealJS):** `macros-header.html` is a small loader script (not a second macro-definition file). It reads `macros.qmd`, converts macro definitions to a MathJax config object, and sets `MathJax.tex.macros` before typesetting. For RevealJS, it also forces the built-in Reveal math plugin to use MathJax 3. This requires `macros.qmd` to be published as a static resource in the site output. [View the RevealJS slides output.](#)

**Note (PDF):** For PDF, `macros.qmd` can be included directly since its raw LaTeX goes into the document preamble.

**Note:** This demo uses `macros-header.html` / `macros.qmd` because both files are at the root of this repository. When using this repository as a Git submodule (cloned into a `macros/` subdirectory), replace with `macros/macros-header.html` and `macros/macros.qmd`.

The same configuration can be placed in a shared `_quarto.yml` to apply project-wide:

```
format:
  html:
    include-in-header:
      - macros/macros-header.html
  pdf:
    include-in-header:
      - macros/macros.qmd
  revealjs:
    include-in-header:
      - macros/macros-header.html
```

This inserts the macro definitions for each output format:

- **PDF:** `macros.qmd` goes into the LaTeX preamble via `include-in-header` — all macros work, including `\providecommand`.
- **HTML:** `macros-header.html` is inserted via `include-in-header` and loads macros from `macros.qmd` into `MathJax.tex.macros`.
- **RevealJS:** `macros-header.html` does the same loader step and switches the Reveal math plugin to MathJax 3, so slide math uses the same `macros.qmd` source.
- **docx:** Word documents do not use a LaTeX preamble or MathJax header, so `include-in-header` has no effect on math rendering for docx. The `docx: default` entry above produces docx output without macro support.

## Example math

The table below uses macros defined with `\def` in `macros.qmd`, which work in all output formats that support MathJax or LaTeX:

| Description     | Code                                                    | Output                                   |
|-----------------|---------------------------------------------------------|------------------------------------------|
| Greek: alpha    | <code>\a</code>                                         | $\alpha$                                 |
| Greek: beta     | <code>\b</code>                                         | $\beta$                                  |
| Greek: gamma    | <code>\g</code>                                         | $\gamma$                                 |
| Greek: lambda   | <code>\l</code>                                         | $\lambda$                                |
| Greek: sigma    | <code>\s</code>                                         | $\sigma$                                 |
| Log-likelihood  | <code>\llik</code>                                      | $\ell$                                   |
| Independent     | <code>X \ind Y</code>                                   | $X \perp\!\!\!\perp Y$                   |
| IID distributed | <code>X_i \siid \Normal(\m,</code><br><code>\ss)</code> | $X_i \sim_{\text{iid}} N(\mu, \sigma^2)$ |
| Vector: mu      | <code>\vm</code>                                        | $\tilde{\mu}$                            |

| Description | Code             | Output        |
|-------------|------------------|---------------|
| Hat beta    | <code>\hb</code> | $\hat{\beta}$ |

## Color macros

`macros.qmd` defines a one-argument macro for each of the 19 [LaTeX base colors](#) (`\red`, `\blue`, `\green`, `\orange`, ...), using the `\color` switch form (`{\color{red}{#1}}`), which is supported by MathJax (HTML/RevealJS) and lualatex (PDF):

| Description   | Code                                                                           | Output                                                                                                               |
|---------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Red           | <code>\red{x}</code>                                                           | $x$                                                                                                                  |
| Blue          | <code>\blue{y}</code>                                                          | $y$                                                                                                                  |
| Green         | <code>\green{z}</code>                                                         | $z$                                                                                                                  |
| Orange        | <code>\orange{\mu}</code>                                                      | $\mu$                                                                                                                |
| Combined      | <code>\red{x} + \blue{y} = \green{z}</code>                                    | $x + y = z$                                                                                                          |
| Nested macros | <code>X_i \sim_{\red{\ind}} \blue{\Normal}(\orange{\mu_i}, \green{\ss})</code> | $X_i \sim_{\textcolor{red}{\text{ind}}} \textcolor{blue}{N}(\textcolor{orange}{\mu_i}, \textcolor{green}{\sigma^2})$ |

**Note (docx):** as described above, the `include-in-header` strategy does not load macros for docx output, so the color macros (like all other macros) are unavailable there. For docx, use the [include shortcode strategy](#) instead.